Sweatshop on P-Chess: Stockfish analysis

Stockfish 19 shipped (CI-004), a second engine on the UCI bridge (REV-001), game analysis (REV-002), move classification (REV-003) and accuracy (REV-004); GPT-6.1 Terra lineup, lane `analysis`

Foreman, session `sweatshop/analysis/2026-10-30-1831`

30 October 2026

Summary

A lane run (`-Lane analysis`) in a fresh clone at `.artifacts-build/lanes/analysis`, default lineup `terra`: stage 2 on `gpt-6.1-terra high` for the two `normal` tickets and `gpt-6.1-terra xhigh` for the three `hard` ones; review on `gpt-6.1-sol xhigh`. The lanes `repertoire` and `v010` ran at the same time in their own clones. Launch at 18:31, relaunch at 19:03 after a proxy answer, the five planned tickets merged by 20:08; a third launch at 20:28 ran a follow-up (CLEAN-009) for the session PR's red Linux CI, merged at 20:36.

- **Produced:** CI-004, REV-001, REV-002, REV-003, REV-004 and the follow-up CLEAN-009 merged on the session, all `Verdict: Approve` (CI-004 and REV-003 `Review: human`, the others `Review: agent`). Session PR #26, checks green (Linux `test` 306/0, `windows-export`). Gate 306 passed, 0 failed, 0 skips (270 on `main` at launch).
- **Cost: $6.31** at API list price over the 8 priced stages: $6.15 for the 7 `gpt-6.1-sol xhigh` reviews, $0.16 for CLEAN-009's `gpt-6.1-terra high` stage (the driver priced Terra from its 20:28 launch); the other 9 Terra stages log `?`. Codex quota: 7 points of the weekly window (Terra high 1, Terra xhigh 3, Sol xhigh 3, outside stages 0; each ±1), shared with the two concurrent lanes.
- **What went wrong:** CI-004 burned two stage-2 attempts and was parked `blocked`. The first ran the gate with a Godot it downloaded itself, after failing to find the installed one; Godot could not write `user://`, child processes multiplied, the gate was killed. The second stopped in a minute, reading the saved attempt as an interrupted `implementing`. The foreman put it to the proxy; the third attempt took 3 minutes. REV-003 was reopened once with three findings (two P2). And the session PR's Linux `test` check was red from REV-001's merge to CLEAN-009's: two bridge tests faked a live engine with the test's own pid, which Linux rejects. The Windows gate and both reviews passed them; the foreman only looked at the PR's checks after the five tickets merged.
- **Provisional verdict:** on real feature work Terra was fast (median 6 min per to-review stage 2) and disciplined (red→green on every behaviour, two mutation checks), and it asked the right question on REV-002. It struggled when the environment did not behave and no instruction covered it, and its test fakes were not portable. Its pair with Sol xhigh as reviewer now has 12 reviews for Terra high (25% reopened) and 5 for Terra xhigh.

What was done

CI-004: `scripts/fetch-engine.ps1` now also downloads the pinned official Stockfish 19 Windows build, checks the SHA-256 of the archive and of the installed exe, and puts `engine/stockfish/stockfish.exe` and its `Copying.txt` beside Lc0; a second run downloads nothing. The release zip ships both, and `docs/THIRD_PARTY.md` gains a Stockfish section. Not reopened. The two failed attempts were environment and state handling, not code (see Driver and runtime issues).

REV-001: the UCI bridge (`engine/uci.gd`) takes a configurable engine and path, so a second engine can run beside Maia's Lc0, and forwards every search `info` line and `bestmove` to the caller. Maia's behaviour is unchanged. Merged with no finding.

REV-002: a new `engine/analyser.gd` runs Stockfish over every position of a saved game, last to first, MultiPV 2, 3 000 000 nodes per position with an early stop once the best move and score settle, and writes `user://analysis/<game>.json` with both lines, the best move, terminal scores, the engine version and the budget; a file from another version or budget reads as stale. `scripts/analyse.gd` does it headless for one PGN. The proxy approved two small additions to the bridge for it (the handshake's `id name`, and `stop` during a search). Merged with no finding. The foreman's probe on one of Bruno's saved games (124 positions) took 21.4 s.

REV-003: a new `core/classification.gd` grades each move (Forced, Repertoire, Book, Brilliant, Great, Best, Excellent, Good, Inaccuracy, Mistake, Blunder, plus Miss, the lenient Blunder and mate rules), with every threshold in `core/tunables.gd`. Reopened once: a capture right after a pawn push was taken for a recapture (only the destination squares were compared), so it lost Great; a free en passant capture got Great because the empty destination square hid the victim; and one 0 cp cutoff stayed a literal outside `core/tunables.gd`. All three in `core/classification.gd`, all fixed test-first in one pass.

REV-004: a new `core/accuracy.gd` gives each move's accuracy and each side's game accuracy with Lichess's published formula. Tested against two real Lichess games, matching the site's figures (87/94 and 85/72). Merged with no finding.

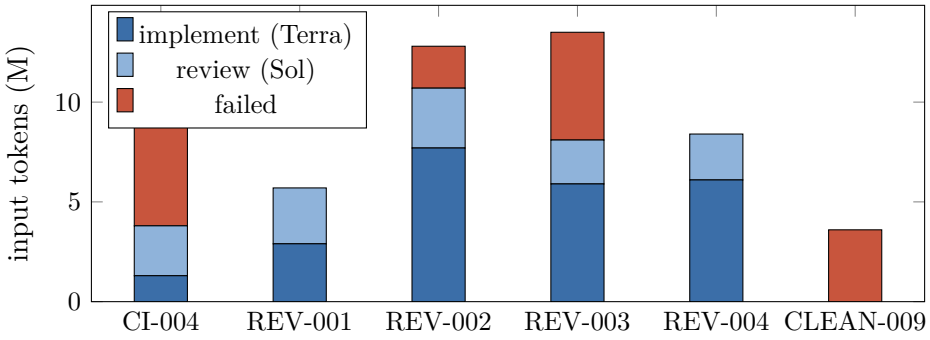
CLEAN-009 (follow-up written by the foreman, triaged by the proxy): nothing changes for the player. Three bridge tests in `tests/test_uci.gd` pretended an engine was running by giving the bridge the test's own process id. Windows accepts that; Linux only tracks its own child processes, so the bridge saw a stopped engine and two tests failed on CI from REV-001's merge on. The tests now start a real idle Godot child (`tests/fixtures/idle_child.gd`), as the runner's own tests already did. Proven red on Windows by mutating the bridge, green on Linux CI (306/0).

Timeline

18:29	Clone of `origin` at `eea8c16`; gate in the clone 270/0. `main`'s CI on `eea8c16` still running (later green).
18:31	Launch, pid 11396. Codex sessions cannot reach the proxy (the driver asks for them).
18:46	CI-004 attempt 1 fails (14 min): code done, gate killed at 198/72 with a downloaded Godot.
18:47	CI-004 attempt 2 stops in 1 min; parked `blocked` after two attempts. REV-001 stage 2.
18:53	REV-001 to review (Terra high, 6 min).
18:59	REV-001 merged, 0 findings. Nothing runnable (REV-002 needs CI-004); PR #26 opened.
19:00	Foreman asks the proxy about CI-004: resume from attempt 1's commit; `c748352`.
19:03	Relaunch, pid 22892; Codex sessions now reach the proxy directly.
19:07	CI-004 to review (Terra high, 3 min).
19:12	CI-004 merged, 0 findings. REV-002 stage 2 (Terra xhigh).
19:17	REV-002 stops to ask (scope of `engine/uci.gd`); its own proxy call fails; the driver asks.
19:19	Proxy decides: two seams on the bridge. REV-002 stage 2 again.
19:30	REV-002 to review (11 min).
19:35	REV-002 merged, 0 findings. REV-003 stage 2.
19:44	REV-003 to review (9 min).
19:48	REV-003 reopened, 3 findings. Stage 2 again.
19:52	REV-003 to review (4 min).
19:56	REV-003 merged. REV-004 stage 2.
20:04	REV-004 to review (8 min); Codex then ends on "model at capacity", after the handoff commit.
20:08	REV-004 merged; nothing left in the lane; PR #26 updated (5 tickets).
20:19	Foreman finds PR #26's Linux `test` check red since 7940566 (304/2, `tests/test_uci.gd`).
20:23	Proxy triages the foreman's follow-up CLEAN-009: tests only, a real child process.
20:28	Third launch, pid 26744 (driver now prices Terra).
20:32	CLEAN-009 to review (Terra high, 4 min).
20:36	CLEAN-009 merged, 0 findings; PR #26 updated (6 tickets).
20:38	PR #26's Linux `test` check green, 306/0.

Per stage

When	Ticket	Stage	Model	Min	In	Cache	Out	Cost	Outcome
18:46	CI-004	impl	terra high	14	5.4M	98%	12k	?	failed
18:47	CI-004	impl	terra high	1	0.2M	88%	1k	?	blocked
18:53	REV-001	impl	terra high	6	2.9M	97%	9k	?	to-review
18:59	REV-001	review	sol xhigh	6	2.8M	95%	15k	0.95	merged
19:07	CI-004	impl	terra high	3	1.3M	96%	3k	?	to-review
19:12	CI-004	review	sol xhigh	5	2.5M	96%	13k	0.84	merged
19:17	REV-002	impl	terra xhigh	5	2.1M	97%	6k	?	asked
19:30	REV-002	impl	terra xhigh	11	7.7M	99%	26k	?	to-review
19:35	REV-002	review	sol xhigh	4	3.0M	94%	14k	1.04	merged
19:44	REV-003	impl	terra xhigh	9	5.9M	98%	25k	?	to-review
19:48	REV-003	review	sol xhigh	4	3.1M	95%	15k	1.07	reopened
19:52	REV-003	impl	terra xhigh	4	2.3M	97%	7k	?	to-review
19:56	REV-003	review	sol xhigh	4	2.2M	93%	11k	0.85	merged
20:04	REV-004	impl	terra xhigh	8	6.1M	98%	16k	?	to-review
20:08	REV-004	review	sol xhigh	4	2.3M	94%	13k	0.85	merged
20:32	CLEAN-009	impl	terra high	4	1.6M	97%	6k	0.16	to-review
20:36	CLEAN-009	review	sol xhigh	4	2.0M	94%	12k	0.55	merged
				96	53.4M		204k	6.31+?	



The chart is input tokens, not dollars: most Terra stages are unpriced. Red: CI-004's two failed attempts (5.6M; attempt 1's code was reused, so only its gate run and attempt 2 were lost), REV-002's asking attempt (2.1M, a legitimate question, not a defect), REV-003's first review plus fix round (3.1M + 2.3M), and all of CLEAN-009 (1.6M + 2.0M, the repair of REV-001's and REV-002's non-portable tests).

Comparison with previous runs

From `run-log.md`: over its P-Chess runs, `gpt-6.1-sol high` took a median of about 11 minutes per stage 2 and `gpt-6.1-sol xhigh` 11–22 on hard tickets, at $0.4–1.5 per stage. This run's Terra stage 2s that reached review took 3–11 minutes (median 6): 6, 3 and 4 on the normal tickets, 11, 9 and 8 on the hard ones. Reviews cost what they did before ($0.55–1.07 against $0.75–1.26). The repertoire and v010 lanes, run in parallel on the same lineup, saw the same speed and a related weak spot: tickets stalled on environment or handoff hygiene, not on the code.

Observations per model

- **gpt-6.1-terra high, stage 2** (5 stages, CI-004, REV-001 and CLEAN-009): REV-001 looked clean (red, green, a second red/green for search forwarding, a mutation run, gate 274/0 in 6 minutes), but its fake engine used the test's own pid, which only Windows accepts; REV-002 copied the pattern. Only Terra commits use it (`git log -S "OS.get_process_id()" - tests/`); Sol's earlier bridge tests ran a real Lc0 and skipped without it. CLEAN-009, with the proxy's recipe, fixed it in 4 minutes with the mutation reds the ticket asked for. CI-004's code was right the first time, but the stage handled the environment badly: it did not find `godot` (REV-001's session, in the same sandbox, found it through `where.exe` and the profile), downloaded Godot from GitHub instead, and then read the global "re-run the gate only after a code change" rule as "the gate cannot run again". Attempt 2 refused to start because the saved attempt's ticket text said `implementing`, although the session's copy said `to-implement`. With the proxy's explicit recipe the third attempt was flawless.
- **gpt-6.1-terra xhigh, stage 2** (5 stages, the hard tickets): stopped at the right place on REV-002 (a seam the ticket did not name), with a precise question; then 11 alternating test/production commits, +574 lines in scope. REV-003 and REV-004: one `test:` commit before every behaviour, a self-corrected test expectation verified red first, a mutation check on REV-004, and real Lichess games as fixtures. Missed: chess edge cases in its own exclusion logic (recapture, en passant) and one threshold left out of the tunables file. On REV-003 it noticed that criterion 7's "+600 cp after" branch cannot be reached under the clamped formula and documented it instead of asking; the reviewer agreed. One commit-hygiene slip: the P3 fix rode in the en passant commit.
- **gpt-6.1-sol xhigh, review** (7 stages, 4–6 min): verified receipts and red/green order, re-checked CI-004's hashes against the official release, and reproduced both REV-003 P2s with an external probe. The re-review raised nothing new (no drip-feed). No production code written in review, so no round-0 rows. Its 4-minute pass over REV-002's +574 lines found nothing; the foreman's real-game probe agrees with it. It reused the Windows gate receipt every time and never looked

at the session PR's Linux check, so it approved REV-002 while that check was already red (review miss on the copied fake-pid test). CLEAN-009's review flagged the Linux confirmation as pending for the foreman.

Driver and runtime issues

1. `$CodexPrices` had no `gpt-6.1-terra` row for the first two launches: 9 Terra stages log `?`. The driver at the 20:28 launch (SHA256 `5C16FF...`, changed locally between launches) priced CLEAN-009's stage at $0.157.
2. The session PR's Linux `test` check was red from 7940566 (REV-001's close, about 19:00) until aad8460 (20:38). The driver pushes the session and moves on; no stage and no review reads the PR's checks, and the foreman only looked after the five tickets merged. Every gate in the loop runs on Windows, CI runs on Ubuntu.
3. CI-004's sandbox gate: `make_dir_recursive` failures under `user://` with a Godot downloaded into `%TEMP%`, runner children timing out and spawning more Godots. The stage killed its own processes and the gate exited −1. The installed Godot passed the full gate in the same sandbox minutes later, so the stage's tool discovery was the cause, not the sandbox alone.
4. Proxy reachability changed between launches: "unreachable" at 18:32, "reached directly" at 19:04. Yet REV-002's stage still got `ECONNREFUSED`, and Codex's automatic approval refused a networked retry ("could send repository context to an external service"). The driver's fallback (asking for it) worked.
5. REV-004's Codex turn ended with "Selected model is at capacity" (exit 1, no final message), seconds after the handoff commit. The driver read the committed state correctly. Earlier in the stage, it would have cost an attempt.
6. Tests share fixed-name files under `user://` (such as `test_ui006_settings.cfg` and `runner_cases/`) across every clone, since the project name is the same. Concurrent lane gates could collide. Not observed this run.
7. Shutdown 1.0–3.9 s per stage; no crash, no timeout. The driver started with "local edits; not updated" at all three launches.

Recommendations

1. Watch the session PR's checks, not only the local gate: have the driver (or the foreman, at every finished stage) read the PR's `test` check after each push, and park the next ticket when it turns red. A Linux-capable gate (WSL, or a `Sync gate:` that runs the suite under the Linux Godot) would have caught the fake-pid tests at stage 2.
2. Commit the driver change that prices `gpt-6.1-terra` (it ran only as a local edit); without it most Terra stages stay unpriced.
3. Put the gate's Godot in the repo's bindings (or have `gate-evidence.ps1` resolve it) and say "never download a toolchain" in ticket-flow: Terra spent a 14-minute stage on a self-downloaded Godot.
4. In ticket-flow, say that a saved recovery commit's `Stage:` is evidence, not state, and that "re-run the gate only after a code change" does not forbid a first complete run. Both misreadings came from rules, not from the code.
5. For classification-like tickets, list the chess edge cases in the test list (en passant, captures after non-captures, promotions); Terra tested the named exclusions but not their edge cases.
6. Keep Terra as the default for now: scoreboard row `gpt-6.1-terra high` → `gpt-6.1-sol xhigh` has 12 reviews (all three lanes), 3 reopened, S2 rate 25%, against `gpt-6.1-sol high` → `gpt-6.1-sol xhigh` 18 reviews, 22% — comparable. `gpt-6.1-terra xhigh` → `gpt-6.1-sol xhigh` 5 reviews, 1 reopened, 20% (an anecdote) against `gpt-6.1-sol xhigh` → `gpt-6.1-sol xhigh` 9 reviews, 11%. The reopen rate does not count the two session-review findings, nor CI-004's two failed attempts and the repertoire lane's parked handoff; those are where Terra trails Sol.

Human debt and proxy decisions

No `Débito humano:` lines. Three `Proxy decided` lines, all by `opus-5.5 xhigh` with receipts:

- **CI-004** (asked by the foreman): "Accept with one change: set CI-004 to `to-implement` and resume from `recovery/20261030-1847/CI-004` by cherry-picking only 423f2ed (drop aa2f15c entirely), then redo the manual check and run the gate once with the installed Godot – the code is done and the only problems were the environment and the stale ticket text."
- **REV-002** (asked by the driver): "Approve widening `engine/uci.gd` and `tests/test_uci.gd` by exactly two seams: the handshake's `id name`, and an in-search `stop`. – Criteria 2, 5 and 6 can't be met without them. Spec ğ15.3 says Stockfish runs 'through the same bridge as Lc0', and the REV-001 handoff already left these two paths to REV-002."
- **CLEAN-009** (stage-1 triage asked by the foreman): "Approve CLEAN-009 as written and run it now in the analysis session, at Difficulty normal with Review agent. Fix the tests only, by using a real child process; one engine/uci.gd seam is pre-approved as a fallback. The red proof comes from mutating production code on Windows, with CI run 41143449186 cited as the cause." Stage 2 did not need the fallback seam.

Reopen attribution

Labelled by the foreman (opus-5.5); the reviewer was `gpt-6.1-sol xhigh`, so no self-judging. Rows appended to `.scratch/reopen-attribution.md`.

Ticket (round)	Finding	Label	Why
REV-003 (1)	A capture after a pawn push taken for a recapture; loses Great	S2-explícito/ código	criterion 5 names the recapture exclusion
REV-003 (1)	Free en passant capture gets Great (victim not on the destination square)	S2-explícito/ código	criterion 5, free-piece exclusion
REV-003 (1)	0 cp mate cutoff left as a literal	S2-explícito/ código	criterion 9 (P3)
After approval, discovery `session-review` (PR #26's Linux `test` check)			
REV-001 (–)	Bridge test fakes a live engine with the test's own pid; fails on Linux	S2-implícito ?	CI runs on Ubuntu and `tests/test_runner.gd` already uses a real child; review miss unknown (CI not visible yet)
REV-002 (–)	Second test copies the same fake; fails on Linux	S2-implícito ?	same cause; review miss yes (PR check already red at the review)

Ticket review totals: S2-explícito/código 3, S2-explícito/teste 0, S2-implícito 0, S1 0, S3-ruído 0; no review miss, no review-induced churn, no fix made in review. Session review (counted apart, not in the reopen rate): 2 findings, both S2-implícito ?, origins REV-001 `513acab` and REV-002 `8a79621`, both fixed by CLEAN-009 (`e40f327`, CI run 41145544373 306/0); one review miss, one unknown. Copilot's review of PR #26 did not run (quota). No release-audit or post-release finding.

Scoreboard

`scoreboard.ps1` over the ClassyCall, EasyMath, PhysiSyst, SIGAA-ME and SynchroNice main checkouts, plus this lane's clone in place of P-Chess's main checkout (same merged run log; the clone holds this run's attribution rows, which reach `main` with this report). It now holds all three lanes of 30 October. Moved since the repertoire report (the v010 report came out between this report's two versions): `gpt-6.1-terra high` 4 → 15 stage runs (1 → 3 failed: CI-004 twice, OPEN-006 once; CLEAN-009 the first priced Terra stage, $0.16), its pair with Sol xhigh 3 → 12 reviews, 3 reopened, S2 rate 33% → 25%; new rows `gpt-6.1-terra xhigh` (6 runs, 5 to review, 1 asked, median 9 min)

and its pair (5 reviews, 1 reopened, S2 20%); session-review findings 28 → 30; `proxy-decision (opus-5.5 xhigh)` 4 (this lane 3, the others 1). The other rows only changed order.

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| Implementer | Stage runs | To review | Failed | Asked | Median min to review | Cost |
|---|---|---|---|---|---|---|
| gpt-6.1-sol high | 64 | 59 | 0 | 5 | 9m | $49.27 (64/64 priced) |
| opus-5.5 medium | 3 | 2 | 1 | 0 | 6m | $4.79 (3/3 priced) |
| opus-5.5 | 7 | 4 | 0 | 3 | 14m | $15.12 (7/7 priced) |
| gpt-6-luna max | 89 | 61 | 10 | 18 | 17m | $4.91 (85/89 priced) |
| sonnet-5.5 high | 2 | 2 | 0 | 0 | 7m | $3.77 (2/2 priced) |
| gpt-6-astra high | 1 | 0 | 0 | 1 | - | $3.92 (1/1 priced) |
| opus-5.5 xhigh | 18 | 14 | 4 | 0 | 12m | $91.78 (18/18 priced) |
| gpt-6.1-sol max | 6 | 5 | 0 | 1 | 24m | $12.12 (6/6 priced) |
| haiku-5.5 high | 2 | 0 | 2 | 0 | - | $1.00 (2/2 priced) |
| opus-5.5 high | 20 | 13 | 6 | 1 | 6m | $37.46 (20/20 priced) |
| gpt-6.1-sol xhigh | 9 | 9 | 0 | 0 | 12m | $9.84 (9/9 priced) |
| gpt-6.1-terra high | 15 | 12 | 3 | 0 | 7m | $0.16 (1/15 priced) |
| gpt-6.1-terra xhigh | 6 | 5 | 0 | 1 | 9m | ? |
| gpt-6-sol xhigh | 1 | 1 | 0 | 0 | 9m | $0.40 (1/1 priced) |
| fable | 1 | 1 | 0 | 0 | 11m | ? |
| gpt-6-luna | 8 | 4 | 4 | 0 | 37m | ? |
| gpt-6-sol | 6 | 5 | 1 | 0 | 17m | ? |
| gpt-5.6-sol | 5 | 4 | 1 | 0 | 17m | ? |
| gpt-6-astra | 18 | 11 | 6 | 1 | 8m | ? |
| sonnet | 5 | 3 | 2 | 0 | 14m | $10.53 (4/5 priced) |
| sonnet-5 xhigh | 11 | 6 | 2 | 3 | 8m | $18.09 (11/11 priced) |

| Implementer | Reviewer | Reviews | Reopened | Reopen rate | Reopened for S2 | S2 rate | Unclassified | Fixed in review for S2 |
|---|---|---|---|---|---|---|---|---|
| gpt-6.1-sol high | gpt-6.1-sol max | 38 | 12 | 32% | 9 | >=24% | 2 | 1 |
| opus-5.5 medium | gpt-6.1-sol max | 2 | 1 | 50% | 0 | >=0% | 1 | 0 |
| opus-5.5 | opus-5.5 | 4 | 0 | 0% | 0 | 0% | 0 | 0 |
| gpt-6-luna max | opus-5.5 high | 18 | 1 | 6% | 1 | 6% | 0 | 0 |
| gpt-6-luna max | gpt-6-sol max | 43 | 23 | 53% | 22 | 51% | 0 | 0 |
| ? | gpt-6-sol max | 1 | 0 | 0% | 0 | 0% | 0 | 0 |
| sonnet-5.5 high | gpt-6-sol max | 2 | 1 | 50% | 1 | 50% | 0 | 0 |
| opus-5.5 xhigh | gpt-6.1-sol max | 6 | 3 | 50% | 3 | 50% | 0 | 0 |
| ? | gpt-6.1-sol max | 1 | 0 | 0% | 0 | 0% | 0 | 0 |
| gpt-6.1-sol max | gpt-6.1-sol max | 4 | 1 | 25% | 1 | 25% | 0 | 0 |
| ? | opus-5.5 xhigh | 4 | 1 | 25% | 1 | 25% | 0 | 0 |
| opus-5.5 xhigh | opus-5.5 xhigh | 7 | 0 | 0% | 0 | 0% | 0 | 0 |
| opus-5.5 high | opus-5.5 xhigh | 10 | 2 | 20% | 2 | 20% | 0 | 3 |
| gpt-6.1-sol high | gpt-6.1-sol xhigh | 18 | 4 | 22% | 4 | 22% | 0 | 0 |
| gpt-6.1-sol xhigh | gpt-6.1-sol xhigh | 9 | 1 | 11% | 1 | 11% | 0 | 1 |
| gpt-6.1-terra high | gpt-6.1-sol xhigh | 12 | 3 | 25% | 3 | 25% | 0 | 0 |
| ? | gpt-6.1-sol xhigh | 1 | 0 | 0% | 0 | 0% | 0 | 0 |
| gpt-6.1-terra xhigh | gpt-6.1-sol xhigh | 5 | 1 | 20% | 1 | 20% | 0 | 0 |
| gpt-6-sol xhigh | gpt-6-astra high | 1 | 0 | 0% | 0 | 0% | 0 | 0 |
| fable | fable | 1 | 1 | 100% | 0 | >=0% | 1 | 0 |
| ? | opus | 1 | 0 | 0% | 0 | 0% | 0 | 0 |
| gpt-6-luna | gpt-6-sol | 4 | 4 | 100% | 0 | >=0% | 4 | 0 |
| gpt-6-sol | gpt-5.6-sol | 5 | 4 | 80% | 0 | >=0% | 4 | 0 |
| gpt-5.6-sol | gpt-6-astra | 4 | 3 | 75% | 0 | >=0% | 3 | 0 |
| gpt-6-astra | gpt-5.6-sol | 11 | 5 | 45% | 0 | >=0% | 5 | 0 |
| sonnet | opus | 3 | 1 | 33% | 1 | 33% | 0 | 0 |
| sonnet-5 xhigh | opus-5.5 xhigh | 5 | 0 | 0% | 0 | 0% | 0 | 0 |

| Implementer -> Reviewer | Rounds | S2 explicit code | S2 explicit test | S2 implicit | S1 | S3 noise |
|---|---|---|---|---|---|---|
| gpt-6-astra medium -> gpt-6.1-sol max | 8 | 11 | 1 | 4 | 1 | 0 |
| gpt-6.1-sol-fast high -> gpt-6.1-sol-fast max | 2 | 2 | 0 | 0 | 0 | 0 |
| gpt-6.1-sol-fast max -> gpt-6.1-sol-fast max | 6 | 2 | 0 | 3 | 1 | 0 |
| gpt-6.1-sol high -> gpt-6.1-sol max | 26 | 16 | 4 | 23 | 1 | 0 |
| gpt-6-luna max -> opus-5.5 high | 1 | 1 | 1 | 0 | 1 | 0 |
| gpt-6-luna max -> gpt-6-sol max | 23 | 31 | 3 | 23 | 12 | 15 |
| sonnet-5.5 high -> gpt-6-sol max | 1 | 3 | 0 | 2 | 0 | 0 |
| opus-5.5 xhigh -> gpt-6.1-sol max | 3 | 3 | 0 | 2 | 1 | 0 |
| gpt-6.1-sol max -> gpt-6.1-sol max | 5 | 7 | 0 | 5 | 0 | 0 |
| haiku-5.5 high -> gpt-6.1-sol max | 3 | 3 | 1 | 5 | 1 | 0 |
| gpt-6.1-sol high -> gpt-6.1-sol xhigh | 53 | 53 | 4 | 23 | 3 | 0 |
| gpt-6.1-sol max -> gpt-6.1-sol xhigh | 2 | 0 | 0 | 3 | 0 | 0 |
| opus-5.5 high -> opus-5.5 xhigh | 8 | 7 | 0 | 3 | 0 | 0 |
| gpt-6.1-sol xhigh -> gpt-6.1-sol xhigh | 3 | 4 | 0 | 1 | 0 | 0 |
| opus-5.5 high -> gpt-6.1-sol xhigh | 1 | 2 | 0 | 0 | 0 | 0 |
| gpt-6.1-terra high -> gpt-6.1-sol xhigh | 3 | 5 | 0 | 3 | 0 | 0 |
| gpt-6.1-terra xhigh -> gpt-6.1-sol xhigh | 1 | 3 | 0 | 0 | 0 | 0 |
| sonnet-5.5 high -> gpt-6.1-sol max | 3 | 5 | 0 | 2 | 3 | 0 |
| sonnet-5.5 xhigh -> gpt-6.1-sol max | 3 | 7 | 0 | 3 | 2 | 0 |
| sonnet-5.5 high -> opus-5.5 xhigh | 3 | 1 | 2 | 0 | 0 | 0 |
| gpt-6-astra medium -> gpt-6-astra xhigh | 1 | 1 | 0 | 0 | 0 | 0 |
| fable -> ? | 4 | 5 | 1 | 3 | 1 | 0 |
| sonnet -> ? | 10 | 3 | 6 | 9 | 3 | 2 |
| opus -> ? | 1 | 3 | 0 | 0 | 0 | 0 |
| sonnet? -> ? | 11 | 3 | 4 | 12 | 3 | 5 |
| sonnet -> opus | 1 | 1 | 1 | 0 | 0 | 0 |
| gpt-6-astra low -> gpt-6.1-sol max | 1 | 2 | 0 | 1 | 0 | 0 |

Evidence coverage: legacy flags and absent records are unknown, not zero defects or zero interventions.
| Discovery | Recorded findings | Earlier review miss (yes / known) | Review-induced churn (yes / known) |
|---|---|---|---|
| ticket-review | 375 | 5 / 180 | 1 / 182 |
| session-review | 30 | 23 / 25 | 0 / 29 |
| release-audit | 2 | 2 / 2 | 0 / 2 |
Runtime completion to process exit: 265.366s total over 212 measured stages; 334 unmeasured.
| Intervention | Recorded events |
|---|---|
| human-rescue | 30 |
| automatic-recovery | 89 |
| proxy-decision (unknown) | 37 |
| routine-approval | 18 |
| proxy-decision (gpt-6-astra high, fallback) | 58 |
```

proxy-decision (opus)	42
proxy-decision (supervisor)	12
proxy-decision (opus-5.5 xhigh)	4

Acceptance

The lane has no `spec.md` of its own, so no feature scenarios; the tickets' pending product checks are in the PR. One was machine-checkable and the foreman ran it on the integrated session (`8bb6467`): `scripts/fetch-engine.ps1` verified Stockfish 19, then `scripts/analyse.gd` on a copy of Bruno's saved game `2026-10-26T23-21-19` analysed 124 positions in 21.4 s (exit 0): `engine_version` "Stockfish 19", `node_budget` 3 000 000, two lines on 122 positions, one forced position with one line, the final mate as a terminal score, PGN SHA-256 unchanged. Still Bruno's: classifications and accuracy by eye on a real game once the review UI exists (REV-005, REV-006), and the tagged release with Stockfish on a clean machine.

Pending for the human

1. Review and merge session PR #26 (CI-004 and REV-003 are `Review: human`); its checks are green since CLEAN-009.
2. After the merge, tell the `repertoire` and `v010` foremen to sync.
3. Commit the driver's local edit that prices `gpt-6.1-terra`.